

AWS EC2 - Tasks

1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache

If you mean where to find the User Data box in AWS EC2, follow this:

1. Go to AWS Console → EC2
2. Click Launch instance
3. Scroll down the page.
4. Find Advanced details.
5. Click Advanced details to expand it.
6. Scroll down inside that section.
7. You will see User data – optional.
8. Paste this there:

Used commands :

```
#!/bin/bash
```

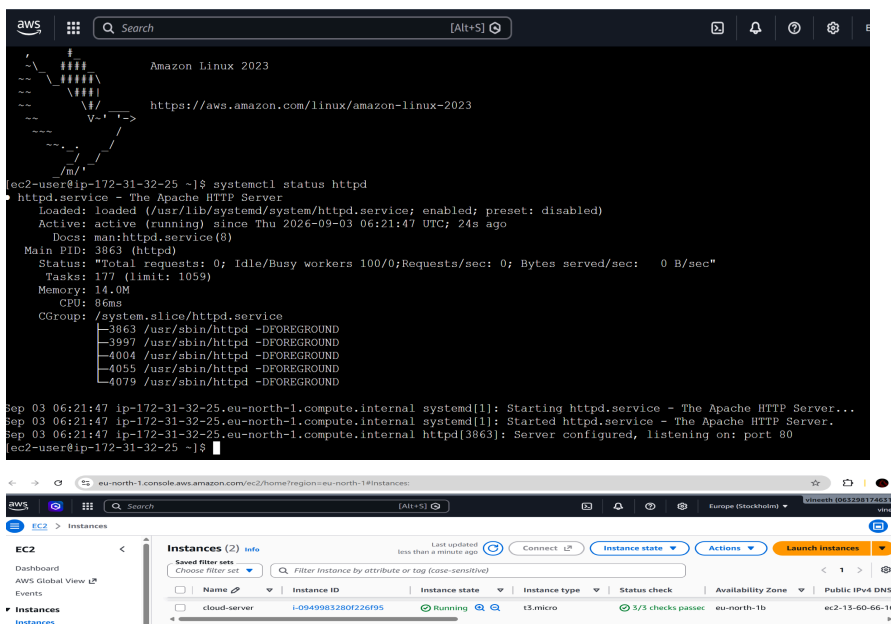
```
yum update -y
```

```
yum install httpd -y
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "Hello from Apache Server" > /var/www/html/index.html
```



2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

Launch Ubuntu EC2 and Install Nginx Using User Data

- 1. Open AWS Console → EC2 → Launch Instance.**
- 2. Enter the instance name as Ubuntu-Nginx.**
- 3. Select Ubuntu Server 24.04 LTS as the AMI.**
- 4. Select the required Instance Type.**
- 5. Select the required Key Pair.**
- 6. Configure the Security Group:**
 - **SSH → Port 22 → My IP**
 - **HTTP → Port 80 → Anywhere (0.0.0.0/0)**
- 7. Enable Auto-assign Public IP.**
- 8. Go to Advanced Details → User Data.**
- 9. Enter the following script:**
- 10. Click Launch Instance.**

```
#!/bin/bash
```

```
yum update -y
```

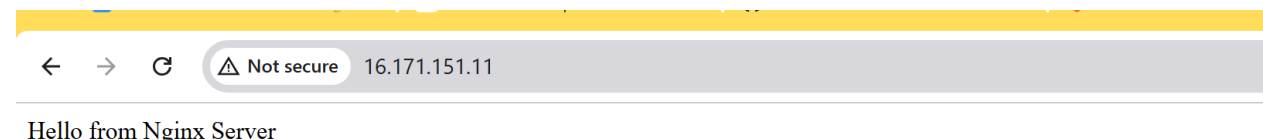
```
yum install httpd -y
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "Hello from Apache Server" > /var/www/html/index.html
```

- 11. Wait until the instance status shows Running.**



```
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-0885196c4aaa63cf2&osUser=ubuntu&sshPort=2...

aws [Alt+S] Ask Amazon Q Europe (Stockholm)

├─773 /usr/sbin/chronyd -n -F 1
├─779 /usr/sbin/chronyd -n -F 1
├─cron.service
├─659 /usr/sbin/cron -f -P
├─dbus.service
├─660 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
├─irqbalance.service
├─670 /usr/sbin/irqbalance
├─multipathd.service
├─180 /usr/sbin/multipathd -d -s
├─networkd-dispatcher.service
├─672 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
├─nginx.service
├─1515 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
├─1517 "nginx: worker process"
├─1518 "nginx: worker process"

ubuntu@ip-172-31-37-76:~$ systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 06:58:24 UTC; 6min ago
     Invocation: 5d49a2a396e249fca25c40aecd620241
       Docs: man:nginx(8).
    Main PID: 1515 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 42ms
    CGroup: /system.slice/nginx.service
            └─1515 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─1517 "nginx: worker process"
                └─1518 "nginx: worker process"

Sep 03 06:58:24 ip-172-31-37-76 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 03 06:58:24 ip-172-31-37-76 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

3.Launch one Windows server and install Tomcat on Windows.

- 1. Open AWS Console → EC2 → Launch Instance.
- 2. Enter the instance name as Windows-Tomcat.
- 3. Select Microsoft Windows Server 2022 Base as the AMI.
- 4. Select the required Instance Type.
- 5. Select the Key Pair.
- 6. Configure the Security Group:
 - Allow RDP → Port 3389 → My IP
 - Allow Custom TCP → Port 8080 → Anywhere
- 7. Enable Auto-assign Public IP.
- 8. Click Launch Instance.
- 9. Wait until the instance status shows Running.
- 10. Select the instance → Connect → RDP Client.
- 11. Get the Administrator password using your key pair.
- 12. Download and open the RDP file to connect to the Windows server.
- 13. Open PowerShell/Command Prompt and check Java:

java -version

- 14. Install Java JDK if Java is not installed.
- 15. Download the Apache Tomcat ZIP file.

```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PowerShellLatest

PS C:\Windows\system32> java - version
Unrecognized option: -
Error: Could not create the Java Virtual Machine.
Error: A fatal exception has occurred. Program will exit.
PS C:\Windows\system32> java -version
java version "1.8.0_503"
Java(TM) SE Runtime Environment (build 1.8.0_503-b01)
Java HotSpot(TM) 64-Bit Server VM (build 25.503-b01, mixed mode)
PS C:\Windows\system32> ^C
PS C:\Windows\system32>
```



4. Take a snapshot of the instance created in Task 2.

Open AWS Console → EC2 → Instances.

Select the EC2 instance created in Task 1.

Go to Actions → Monitor and troubleshoot → Create snapshot.

Select the EBS volume attached to the instance.

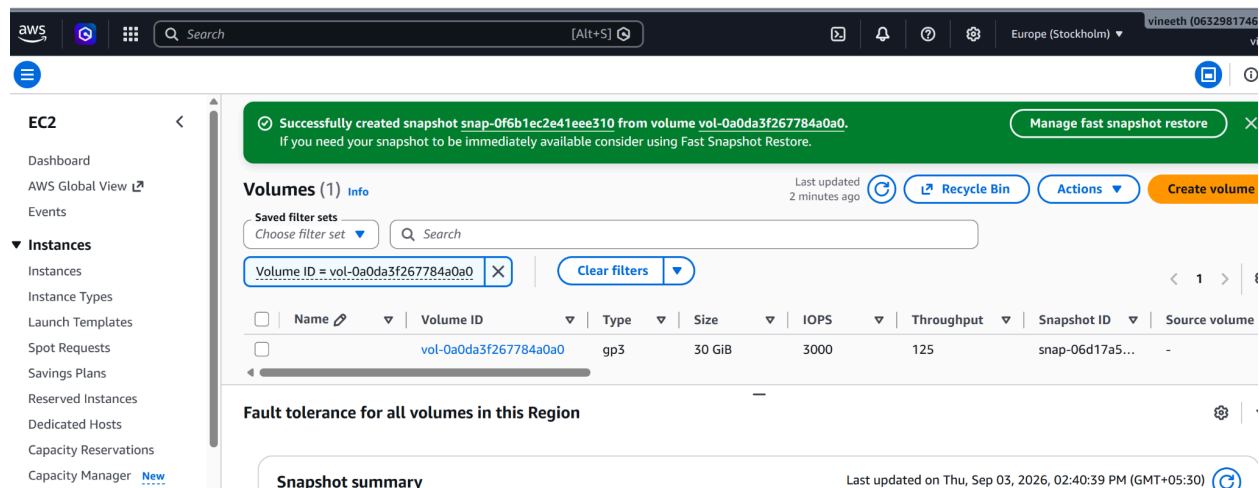
Enter a Description, for example:

Task2 EC2 using Ubuntu image and add a script in user data to install Nginx.

Click Create snapshot.

Go to EC2 → Snapshots.

Verify that the snapshot is created successfully.



5. Assign passwordless authentication for the EC2 created in Task 2.

Steps

1. Open AWS Management Console → EC2.
2. Click Launch instance.
3. Enter an instance name, for example: Spot-Instance
4. Select an available Amazon Linux AMI.
5. Select an instance type, for example: t3.micro
6. Select an existing Key pair.
7. Under Advanced details, find Purchasing option.
8. Select:
Request Spot Instances.
9. Configure the required network and security group settings.
10. Click Launch instance.
11. Go to EC2 → Instances.

13. Verify that the Lifecycle is shown as:

```

Vineeth@LAPTOP-LMH2TOGS MINGW64 /
$ cat /c/Users/Vineeth/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCuwZ8eGjAHg3invwPLBdf/d0TQ58NyjYxTs101JhiofAlD2i+wiT+RCocugKNCr
Fui7h7EiXos/suoxbbjd5s9k161v1k1IdtLp4q88ofw77gk4f2t6TjNqmmD5+/dat01svK6sgoYxvUVR2cu59w/j13kbbd3
RnucH7TosXos/suoxbbjd5s9k161v1k1IdtLp4q88ofw77gk4f2t6TjNqmmD5+/dat01svK6sgoYxvUVR2cu59w/j13kbbd3
7h50260/dRTEbs5Vejjaip6Qtq+UxS6g6Y51tAyx2eXthLWd5sqnrxYDTHcwiX1Tbgv+VRVMeOCHRAFauS0283V/JFEZmYj1RzrF
OVR0V7VR1Z+xl12R16m1Cyobxk+g17RAD9K1zhVf+4X122H911xvsxcWCpYXRiYg+/GjBjdrqgQHM1021aTLueUyLhQ7op2aaVq
h26u5u5AjdCjco/SwvgUx2j+shGhKjcgdxgqPFdUd1f5n2D1Llqurk65/gv1cPxxrLam3IAQoFIVf7VvYb3Gut6NHCs3zqxqXfHv
hKUC11QAHXt6Ak5S+YyhXs8qmxqFJMMjNkBFwBv9Uu/TCkYej1Ruo032jK8Bwgck512cg+IFUFYw0W1kz2t+WP1+Rp2w3c+0K1
Q40Cwsc3UMh1TsFWJ9axQ== Vineeth@LAPTOP-LMH2TOGS

Vineeth@LAPTOP-LMH2TOGS MINGW64 /
$ ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCuwZ8eGjAHg3invwPLBdf/d0TQ58NyjYxTs101JhiofAlD2i+wiT+RCocugK
NcrFui7h7EiXos/suoxbbjd5s9k161v1k1IdtLp4q88ofw77gk4f2t6TjNqmmD5+/dat01svK6sgoYxvUVR2cu59w/j13kbbd3
OSiI02mt0tj1y517vutLsrKfjIGGf5j10pmRf4dZt8q6nbsXqj1P61CMjCq1VfLEKsYd1+19K6brYh1CaIk2ZuX5sW7V506
9u7h7EiXos/suoxbbjd5s9k161v1k1IdtLp4q88ofw77gk4f2t6TjNqmmD5+/dat01svK6sgoYxvUVR2cu59w/j13kbbd3
FEOVR0V7VR1Z+xl12R16m1Cyobxk+g17RAD9K1zhVf+4X122H911xvsxcWCpYXRiYg+/GjBjdrqgQHM1021aTLueUyLhQ7op2aa
Vq26u5u5AjdCjco/SwvgUx2j+shGhKjcgdxgqPFdUd1f5n2D1Llqurk65/gv1cPxxrLam3IAQoFIVf7VvYb3Gut6NHCs3zqxqXfHv
hKUC11QAHXt6Ak5S+YyhXs8qmxqFJMMjNkBFwBv9Uu/TCkYej1Ruo032jK8Bwgck512cg+IFUFYw0W1kz2t+WP1+Rp2w3c+0K1
Q40Cwsc3UMh1TsFWJ9axQ== Vineeth@LAPTOP-LMH2TOGS
Rash: ssh-rsa, com99d, bot f9hdd

```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/Downloads (master)
$ ssh ubuntu@13.49.228.181
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 11:53:12 UTC 2026

System load:  0.0           Temperature:           -273.1 C
Usage of /:   34.9% of 6.61GB Processes:            125
Memory usage: 28%          Users logged in:      0
Swap usage:   0%           IPv4 address for ens5: 172.31.37.76

Expanded Security Maintenance for Applications is not enabled.

150 updates can be applied immediately.
117 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

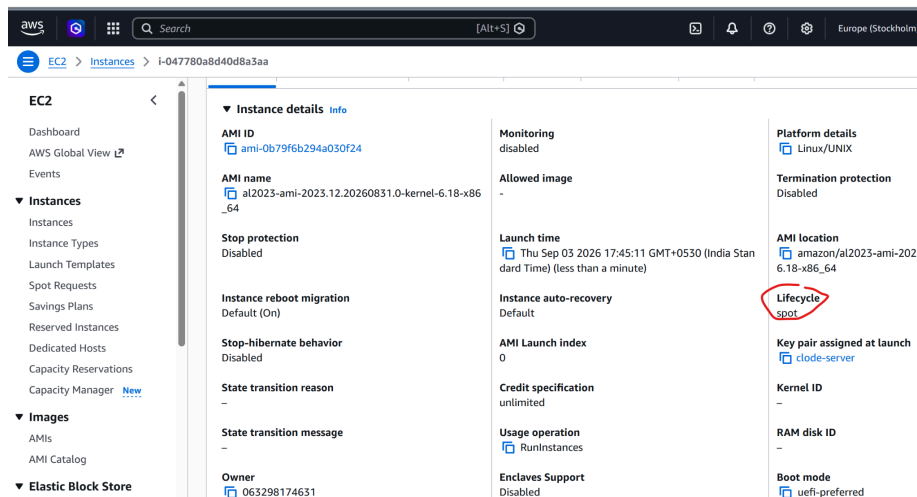
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Sep  3 11:48:52 2026 from 103.160.27.184
ubuntu@ip-172-31-37-76:~$ systemctl status
```

6. Launch any EC2 using the spot purchasing option.

Steps

1. Open **AWS Management Console** → **EC2**.
2. Click **Launch instance**.
3. Enter an instance name, for example: **Spot-Instance**.
4. Select an available **Amazon Linux AMI**.
5. Select an instance type, for example: **t3.micro**.
6. Select an existing **Key pair**.
7. Under **Advanced details**, find **Purchasing option**.
8. Select **Request Spot Instances**.
9. Configure the required **Network and Security Group** settings.
10. Click **Launch instance**.
11. Go to **EC2** → **Instances**.
12. Select the newly launched instance.
13. Verify that the **Lifecycle** is shown as **spot instance**.

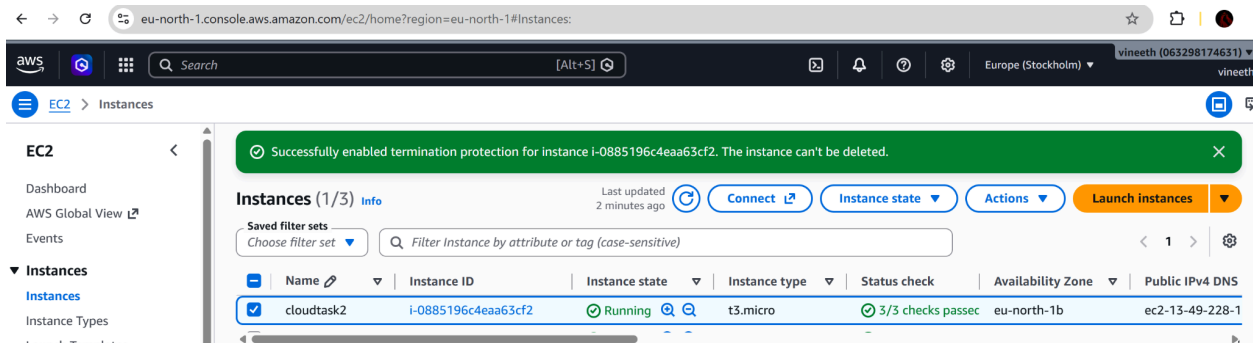


7. Enable termination policy on the EC2 created in Task 2.

Steps

1. Open **AWS Management Console** → **EC2**.
2. Go to **Instances**.
3. Select the **Ubuntu EC2 instance** created in Task 2.

4. Click Actions.
5. Select Instance settings.
6. Click Change termination protection.
7. Select Enable.
8. Click Save.
9. Verify that Termination protection is enabled for the instance



8. Launch one EC2 using AWS CLI.

1. Open Git Bash on your Windows system.
2. Verify AWS CLI installation: `aws --version`
3. Configure AWS CLI: `aws configure` Enter your AWS credentials, region: `ap-south-1` and output format: `json`
4. Verify AWS CLI authentication: `aws sts get-caller-identity`
5. Find an Amazon Linux AMI: `aws ec2 describe-images \`
`--owners amazon \`
`--filters "Name=name,Values=al2023-ami-2023*" \`
`--query "Images | sort_by(@, &CreationDate)[-1].[ImageId,Name]" \`
`--output table`
6. Use the AMI ID obtained from the previous command.
7. Check the available key pairs: `aws ec2 describe-key-pairs --query "KeyPairs[*].KeyName" --output table`
8. Check the available security groups: `aws ec2 describe-security-groups \`
`--query "SecurityGroups[*].[GroupId,GroupName]" \`
`--output table`
9. Launch the EC2 instance using AWS CLI: `aws ec2 run-instances \`
`--image-id ami-0f48d8eda3141f50a \`
`--instance-type t3.micro \`
`--key-name amazon_task \`

**--security-group-ids sg-03e72047d0168c893 **

--count 1

**10. Verify the instance:aws ec2 describe-instances **

**--query "Reservations[].Instances[].[InstanceId,State.Name,InstanceType]" **

--output table

11. Confirm that the instance state becomes:running

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws --version
aws-cli/2.36.38 Python/3.14.6 Windows/11 exe/AMD64

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws configure
AWS Access Key ID [*****]: AKIAQ5PG5P2TSQWQT57T
AWS Secret Access Key [*****]: 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
Default region name [eu-north-1]: eu-north-1
Default output format [json]: json

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws sts get-caller-identity
{
  "UserId": "063298174631",
  "Account": "063298174631",
  "Arn": "arn:aws:iam::063298174631:root"
}

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ AMI:aws ec2 describe-images \
> --owners amazon \
> --filters "Name=name,Values=a12023-ami-2023*" \
> --query "Images | sort_by(@, &CreationDate)[-1].[ImageId,Name]" \
> --output table
bash: AMI:aws: command not found

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 describe-images --owners amazon --filters "Name=name,Values=a12023-ami-2023*"
--query "Images | sort_by(@, &CreationDate)[-1].[ImageId,Name]" --output table
-----
| DescribeImages |
+-----+
| ami-0b79f6b294a030f24 |
| a12023-ami-2023.12.20260831.0-kernel-6.18-x86_64 |
+-----+
```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 describe-key-pairs --query "KeyPairs[*].KeyName" --output table
-----
|DescribeKeyPairs|
+-----+
| c1ode-server |
+-----+

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 describe-security-groups \
> --query "SecurityGroups[*].[GroupId,GroupName]" \
> --output table
-----
|DescribeSecurityGroups|
+-----+
| sg-069777efa12d8f7e0 | launch-wizard-1 |
| sg-0565ebef96d035ba0 | default |
+-----+

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 run-instances \
> --image-id ami-0f48d8eda3141f50a \
> --instance-type t3.micro \
> --key-name amazon_task \
> --security-group-ids sg-03e72047d0168c893 \
> --count 1
```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 run-instances --image-id ami-0b79f6b294a030f24 --instance-type t3.micro --key-name c1ode-server --security-group-ids sg-0565ebef96d035ba0 --count 1
{
  "ReservationId": "r-0b9d406104ac2733c",
  "OwnerId": "063298174631",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "671ec7d0-9819-4216-b26e-124f766235bb",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2026-09-04T07:07:09+00:00",
            "AttachmentId": "eni-attach-0ceaf97c9c0e25d57",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0565ebef96d035ba0",
              "GroupName": "default"
            }
          ],
          "Ipv6Addresses": [],
          "MacAddress": "0a:18:24:87:d8:f5",
          "NetworkInterfaceId": "eni-071c19818be82f844",
          "OwnerId": "063298174631",
          "PrivateDnsName": "ip-172-31-33-41.eu-north-1.compute.internal",
          "PrivateIpAddress": "172.31.33.41",
          "PrivateIpAddresses": [
            {
              "Primary": true,
              "PrivateDnsName": "ip-172-31-33-41.eu-north-1.compute.internal",
              "PrivateIpAddress": "172.31.33.41"
            }
          ],
          "SourceDestCheck": true,
          "Status": "pending"
        }
      ],
      "SubnetId": "subnet-0b79f6b294a030f24",
      "Tags": [],
      "UserData": ""
    }
  ],
  "MaxCount": 1,
  "MinCount": 1
}
```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 describe-instances \
> --query "Reservations[].Instances[].[InstanceId,State.Name,InstanceType]" \
> --output table
```

DescribeInstances		
i-0885196c4eaa63cf2	running	t3.micro
i-0e61f985207796a00	running	t3.micro
i-047780a8d40d8a3aa	running	t3.micro
i-09b7dd7ecb5acd0ce	running	t3.micro